RECEIPT AND INVOICE DIGITIZER

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PROBLEM STATEMENT

- Businesses manage many paper receipts & invoices.
- Manual entry is time-consuming and error-prone.
- Receipts can be lost or damaged.
- Hard to search and analyze expenses.

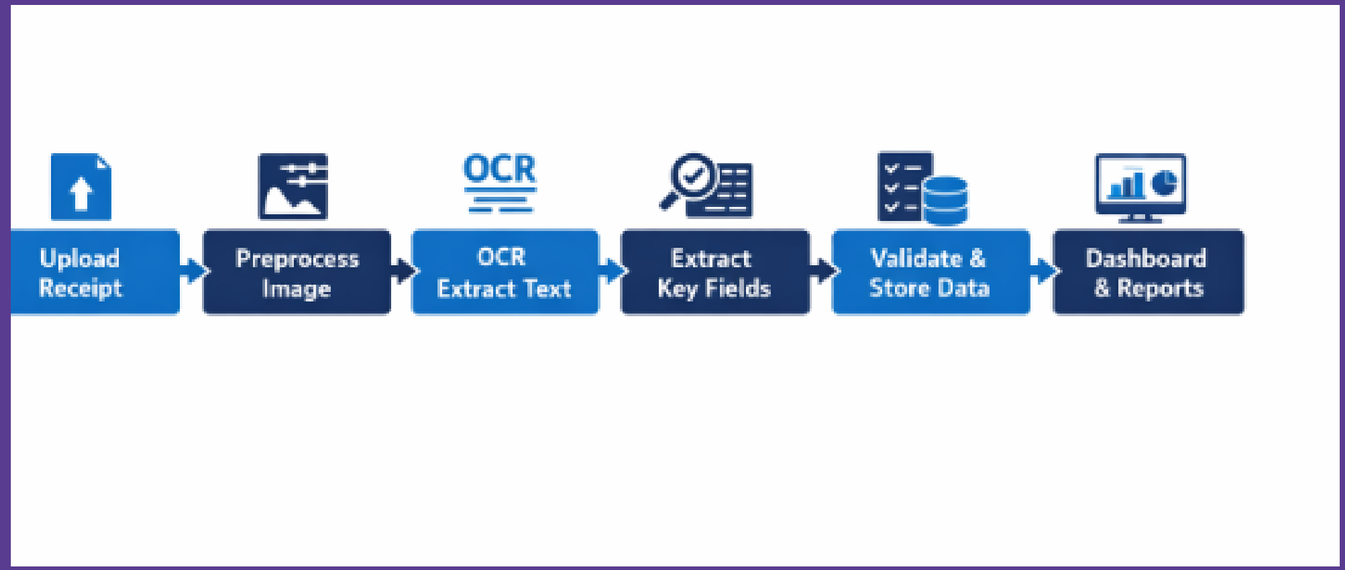
PROPOSED SOLUTION

A system that:

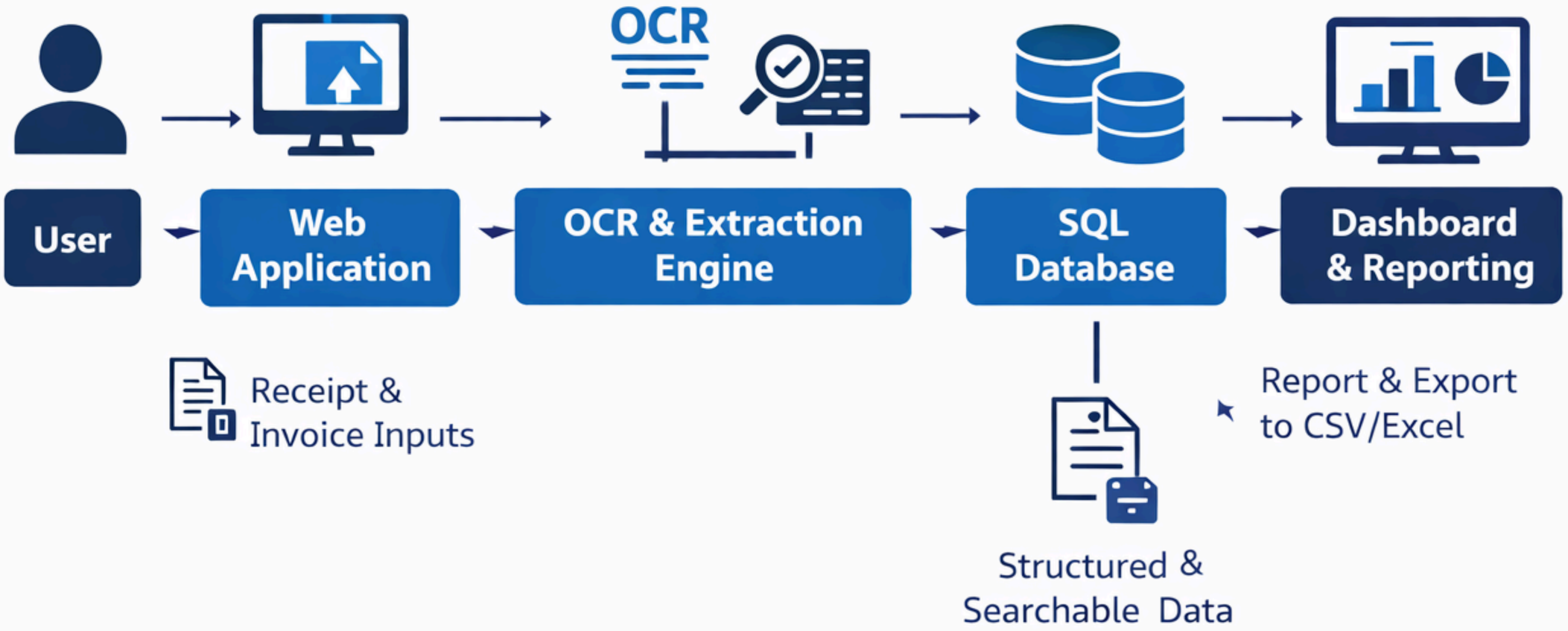
- Scans receipts & invoices
- Extracts text using OCR
- Identifies key data automatically
- Stores digital records
- Provides analytics & reports

PROJECT OBJECTIVES AND WORKFLOW

- Automates data extraction
- Reduces manual effort
- Ensures accurate storage
- Enables quick search & retrieval
- Provides expense insights



System Architecture



MILESTONE-1

Document Ingestion & OCR

Objectives

1. Upload receipts (PDF, JPG, PNG)
2. Preprocess images
3. Extract text using OCR

Techologies Used

1. Python
2. OpenCV / Pillow (image preprocessing)
3. Tesseract OCR / Groq API

Explanation

Receipts are uploaded and enhanced for clarity. OCR converts images into machine-readable text.

MILESTONE-2

FIELD EXTRACTION & VALIDATION

Objectives

- Extract vendor, date, totals, tax
- Validate totals & detect duplicates
- Store structured data

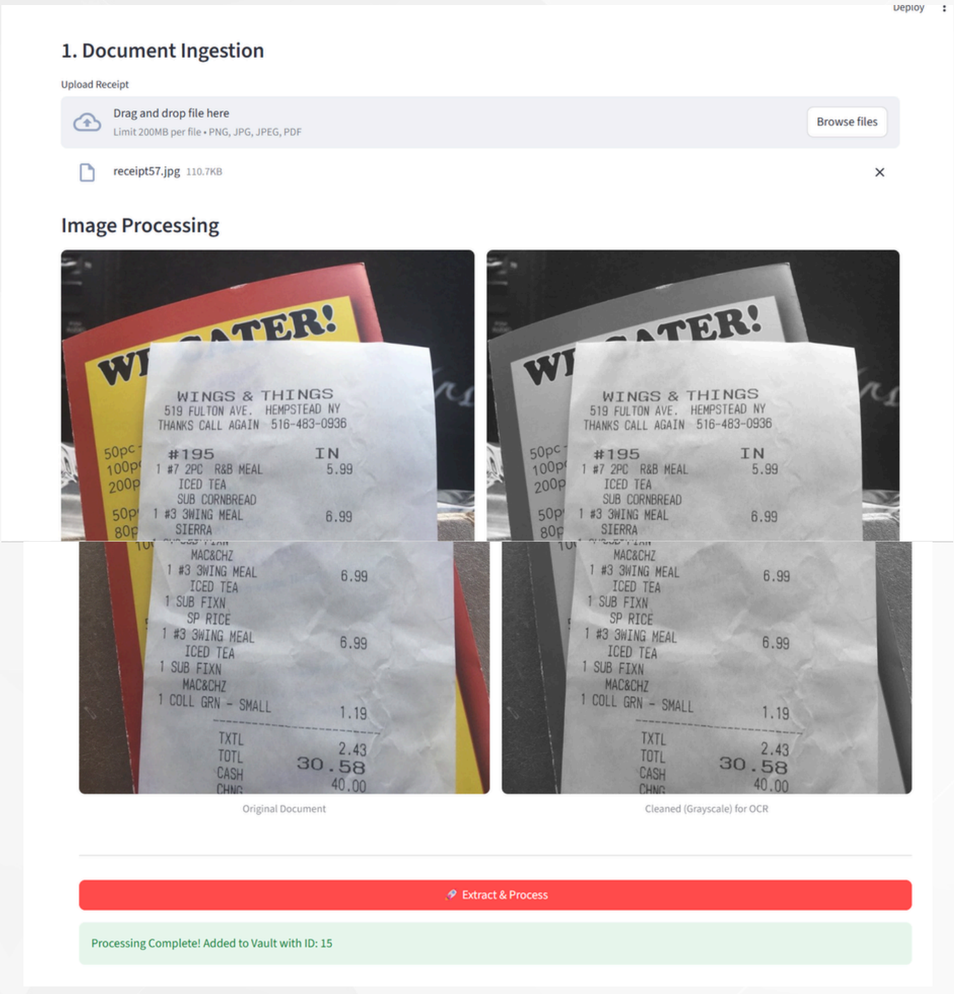
Technologies Used

- Python Regex
- NLP techniques
- Pandas (data structuring)
- Hashing algorithms (duplicate detection)

Explanation

Regex & NLP extract key fields. Validation ensures correct totals and removes duplicates.

Outcomes of Milestone 1 and 2



◆ Vault Status (Recent Entries)

id	merchant	date	total_amount
15	WINGS & THINGS	2026-02-19	90.58
14	WINGS & THINGS	2026-02-19	90.58
10	Kitchen & Pati	2023-09-03	0
5	SOLDEN BOWL TERI	2019-08-18	15.03
13	Tay Ho Oak Land	2019-02-19	16.77

🔍 Validation Details & Database Status

◆ Parsing & Logic Validation

Sum Check: ❌ Invalid: 50.58 + 0.00 != 90.58 (Diff: -40.00)

Duplicate Detected: ✅ No duplicate found

Tax Logic: ✅ Tax Rate: 0.0% (Normal)

Required Fields: ✅ All required fields present

MILESTONE-3

DASHBOARD & REPORTING

Objectives

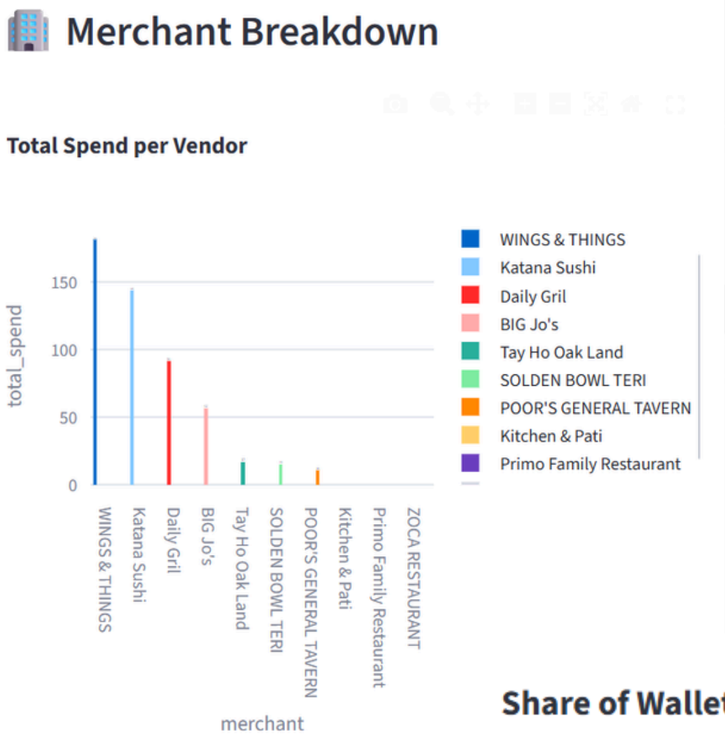
- Build user dashboard
- Export reports (CSV/Excel)
- Show spending analytics

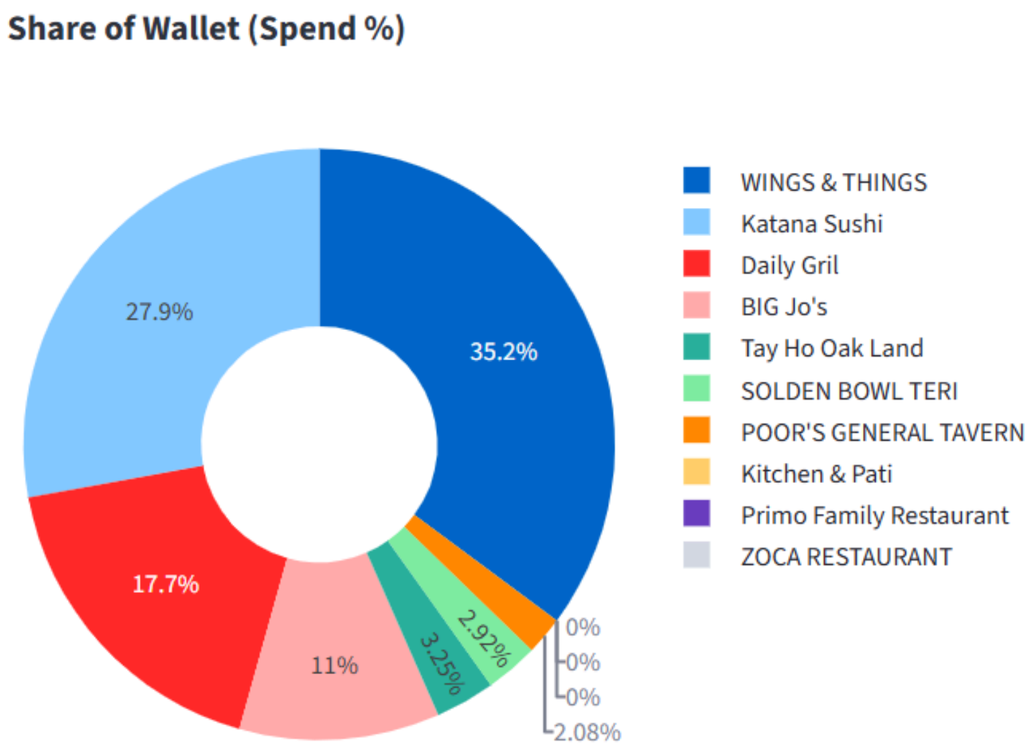
Technologies Used

- Streamlit / Flask
- Pandas
- Matplotlib / Plotly (charts & analytics)

Explanation

Dashboard allows users to review receipts, download reports, and view expense insights.





MILESTONE-4

POLISHING & INTEGRATION

Objectives

- Improve extraction accuracy
- Add search & filter
- Optimize database performance

Technologies Used

- Template-based parsing
- SQL indexing & query optimization
- Caching techniques

Explanation

Template parsing improves accuracy. Search & optimization enhance performance and usability.

CONCLUSION

- The Receipt & Invoice Digitizer simplifies receipt management by turning paper documents into organized digital data.
- It saves time, reduces errors, and provides clear expense insights through an easy-to-use dashboard.
- This project demonstrates how technology can solve real-world problems and improve efficiency.

THANK YOU